

1 Performance Evaluation

Performance is evaluated using two metrics: dominance count and Diversity Comparison Indicator (DCI) [1]. Dominance count measures how many solutions from one algorithm dominate solutions from the other algorithms in the global Pareto front all solutions from all algorithms, while DCI quantifies solution diversity by analyzing their distribution in the objective space. To compare the performance of the discovery. Table 1 summarizes the characteristics of the experiment.

Table 1: Characteristics of experiments

Parameter name	Value
Population Size	20
Iteration	20
Time Limitation	1700
Pareto list size	220
DCI grid size	10
Optimised Model	SDAG
Entropic relevance background model	Z
Event Log	BPI_2013.xes

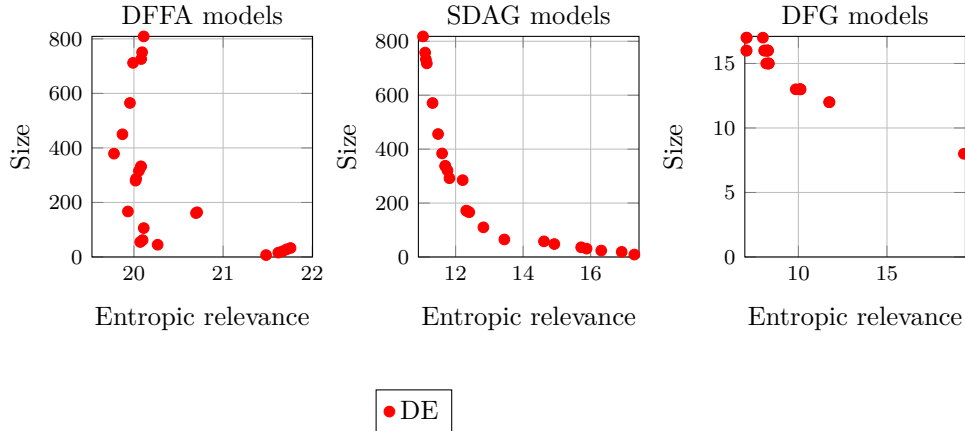


Figure 1: Size and entropic relevance of discovered DFFAs

Table 2: DCI comparison(%)

Event log	DE
BPI_2013.xes	31.56

Tables 2 and 3 summarize the analysis of the diversity of the models and the global Pareto front (dominance count), respectively. Figure 1 shows the characteristics of the Pareto-optimal models discovered from the BPI_2013.xes by the selected algorithm.

Table 3: Pareto dominance count comparison

Event log	DE	Total
BPI_2013.xes	100.00	24 100.00%

Table 4: Execution time comparison (iteration per second)

Event log	DE
BPI_2013.xes	2.14

2 Online Resources

Updated algorithms are publicly available via the following links.

- Process Discovery Optimization.

The following link refers to the details of candidate models generated by algorithms.

- Candidate pareto frontier models.

3 References

[1]. Li, M., Yang, S., Liu, X.: Diversity comparison of Pareto front approximations in many-objective optimiation. IEEE Trans. Cybern.44(12), 2568–2584 (2014)

4 Appendix

Tables 5 lists the quality parameters of candidate models by the selected algorithms.

Table 5: Details of candidates models

Method	size	Entropic relevance
DE	818.0	11.036
DE	321.0	11.760
DE	456.0	11.481
DE	169.0	12.358
DE	384.0	11.601
DE	571.0	11.318
DE	65.0	13.444
DE	733.0	11.119
DE	718.0	11.148
DE	758.0	11.098
DE	110.0	12.824
DE	292.0	11.818
DE	172.0	12.315
DE	338.0	11.693
DE	337.0	11.696
DE	166.0	12.405
DE	9.0	17.294
DE	285.0	12.210
DE	19.0	16.919
DE	24.0	16.314
DE	48.0	14.924
DE	31.0	15.879
DE	58.0	14.617
DE	36.0	15.720